

COMPUTER SCIENCE & IT

DIGITAL LOGIC



BOOLEAN THEOREMS AND GATES -1

Lecture : 08



By- Chandan Gupta Sir



Recap of Previous Lecture

Duality

9 mfs Concepts



Topics to be Covered

Question Practice

[MCQ]



The dual of a Boolean function $F(x_1, x_2, \dots, x_n, +, \cdot, ')$, written as F^D , is the same expression as that of F with $+$ and $\cdot$ swapped. F is said to be self-dual if $F = F^D$.
The number of self-dual functions with n Boolean variables is

[GATE-2022-CS: 1M]

- A $2n$
- B 2^{n-1}
- C 2^{2^n}
- ☒ D $2^{2^{n-1}}$

$$\begin{aligned} n\text{-variables} &\longrightarrow N = 2^n \text{ (terms)} \\ N_B &= 2^N = 2^{2^n} \\ \downarrow \\ S_D &= 2^{2^{n-1}} \end{aligned}$$

[MCQ]



If w, x, y, z are Boolean variables, then which one of the following is **INCORRECT?** [GATE-2017-CS: 1M]

A $wx + w(x + y) + x(x + y) = x + wy$ ✓

B $\overline{w\bar{x}(y + \bar{z}) + \bar{w}x} = \bar{w} + x + \bar{y}z$ ✓ $\rightarrow \frac{\bar{w}}{\bar{w} + x + \bar{y}z + \bar{w}x} = \bar{w} + x + \bar{y}z$

C $(w\bar{x}(y + x\bar{z}) + \bar{w}\bar{x})y = x\bar{y}$ ✗ $w\bar{x}y(y + x\bar{z}) + \bar{w}\bar{x}y = w\bar{x}y + \bar{w}\bar{x}y = \bar{x}y[w + \bar{w}] = \bar{x}y$

D $(w + y)(wxy + wyz) = wxy + wyz$ ✓

$\bar{x}[wy + \bar{w}]y$
 $\bar{x}[wy + \bar{w}y]$
 $\bar{x}y \cdot 1 = \bar{x}y$

$(w + y)wy(x + z)$
 $wy(x + z)$
 $= wxy + wyz$

[MCQ]

The binary operator $\neq$ is defined by the following truth table

P	Q	$p \neq q$
0	0	0
0	1	1
1	0	1
1	1	0

Δ XOR

Which one of the following is true about the binary operator $\neq$?

[GATE-2015-CS: 1M]

- ☒ A Both commutative and associative
- ☐ B Commutative but not associative
- ☐ C Not commutative but associative
- ☐ D Neither commutative nor associative

[MCQ]



Given the function $F = P' + QR$, where F is a function in three Boolean variables P, Q and R and $P' = !P$, consider the following statements.

(S1) $F = \sum(4, 5, 6)$ ✗

(S2) $F = \sum(0, 1, 2, 3, 7)$ ✓✓

(S3) $F = \prod(4, 5, 6)$ ✓✓

(S4) $F = \sum(0, 1, 2, 3, 7)$

$$\overline{P} + QR$$

$$= \sum(0, 1, 2, 3, 7)$$

$$= \prod(4, 5, 6)$$

$$\overline{P}$$

$$0 \ 0 \ 0 \rightarrow 0$$

$$0 \ 0 \ 1 \rightarrow 1$$

$$0 \ 1 \ 0 \rightarrow 2$$

$$0 \ 1 \ 1 \rightarrow 3$$

$$\begin{array}{ccc} & Q & R \\ 0 & 1 & 1 \rightarrow 3 \\ 1 & 1 & 1 \rightarrow 7 \end{array}$$

Which of the following is true?

[GATE-2015-CS: 2M]

☒ A (S1) – False, (S2) – True, (S3) – True, (S4) – False ✓✓

☐ B (S1) – True, (S2) – False, (S3) – False, (S4) – True ✗

☐ C (S1) – False, (S2) – False, (S3) – True, (S4) – True ✗

☐ D (S1) – True, (S2) – True, (S3) – False, (S4) – False ✗

[MCQ]

The truth table

X	Y	F(X, Y)
0	0	0
0	1	0
1	0	1
1	1	1

$$F(X, Y) = \sum(2, 3)$$

$$= X\bar{Y} + XY$$

$$= X \cdot 1 = X$$

represents the Boolean function

- ☒ A x
- ☐ B $x+y$
- ☐ C $x \oplus y$
- ☐ D y

$$= \pi(0, 1) \quad [\text{GATE-2012-CS: 1M}]$$

$$= (x+y)(x+\bar{y})$$

$$= x + y\bar{y} = x$$

[MCQ]

The simplified SOP (sum of product) form of the Boolean expression

$$(P + \bar{Q} + \bar{R}) \cdot (P + \bar{Q} + R) \cdot (P + Q + \bar{R})$$

[GATE-2011-CS: 1M]

$$\begin{aligned} & \left[(P + \bar{Q}) + \bar{R} \cdot R \right] \left[P + Q + \bar{R} \right] \\ & (P + \bar{Q}) (P + Q + \bar{R}) \end{aligned}$$

$$= P + (\bar{Q} + \bar{R}) \cdot (\bar{Q} + R) \cdot (Q + \bar{R})$$

$$\begin{aligned} & = P + \bar{Q} \cdot (Q + \bar{R}) \\ & = P + \bar{Q} \bar{R} \end{aligned}$$

- ☐ A $(\bar{P} \cdot Q + \bar{R})$
- ☒ B $(P + Q \cdot \bar{R})$
- ☐ C $(P' \cdot Q + R)$
- ☐ D $(P \cdot Q + R)$

[MCQ]



The minterm expansion of $f(P, Q, R) = PQ + Q\bar{R} + P\bar{R}$ is

[GATE-2011-CS: 1M]

PQ	$Q\bar{R}$	$P\bar{R}$
$110 \rightarrow 6$	$010 \rightarrow 2$	$100 \rightarrow 4$
$111 \rightarrow 7$	$110 \rightarrow 6$	$110 \rightarrow 6$

- ☒ A $m_2 + m_4 + m_6 + m_7$
- ☐ B $m_0 + m_1 + m_3 + m_5$ ~~X~~
- ☐ C $m_0 + m_1 + m_6 + m_7$ ~~X~~
- ☐ D $m_2 + m_3 + m_4 + m_5$ ~~X~~

[MCQ]



If P, Q, R are Boolean variables, then $(P + Q') (P.Q' + P.R) (P'.R' + Q')$ Simplifies to

$$[\bar{Q} + P.\bar{P}.\bar{R}] [P\bar{Q} + PR] \text{ [GATE-2008-CS: 1M]}$$

$$\bar{Q} [P\bar{Q} + PR]$$

$$P.\bar{Q} [\bar{Q} + R] = P.\bar{Q}$$

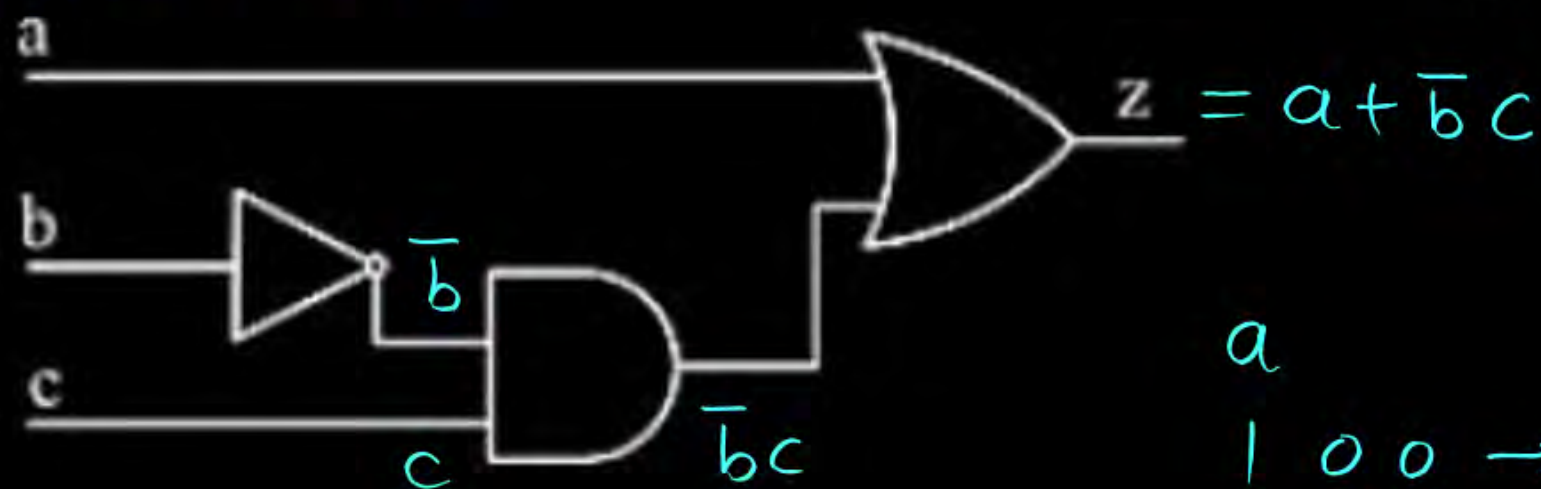
$$P (P + \bar{Q}) (\bar{Q} + R) (\bar{P}\bar{R} + \bar{Q})$$

$$P. [\bar{Q} + 0] = P\bar{Q}$$

- ☒ A $P.\bar{Q}$
- ☐ B $P.\bar{R}$
- ☐ C $P.\bar{Q} + R$
- ☐ D $P.\bar{R} + Q$

[MCQ]

Consider the Boolean function $z(a,b,c)$. Which one of the following minterm lists represents the circuit given above? [GATE-2020-CS: 2M]



a			
1	0	0	$\rightarrow 4$
1	0	1	$\rightarrow 5$
1	1	0	$\rightarrow 6$
1	1	1	$\rightarrow 7$

$\bar{b}$	c	
0	0	$\rightarrow 1$
1	0	$\rightarrow 5$

- ☒ A $Z = \Sigma (0, 1, 3, 7)$ ~~X~~
- ☐ B $Z = \Sigma (2, 4, 5, 6, 7)$ ~~X~~
- ☒ C $Z = \Sigma (1, 4, 5, 6, 7)$
- ☐ D $Z = \Sigma (2, 3, 5)$ ~~X~~

[MCQ]



Which one of the following is NOT a valid identity?

[GATE-2019-CS: 1M]

☒ A $(x + y) \oplus z = x \oplus (y + z)$ ✗ $(x+y)+z$

Handwritten: 0 1 1 0 1 1

☐ B $(x \oplus y) \oplus z = x \oplus (y \oplus z)$ ✓✓

☐ C $x \oplus y = x + y$ if $xy = 0$ ✓✓ $x \oplus y = x + y$

Handwritten: if xy=0 is circled in blue

☐ D $x \oplus y = (xy + x'y')$ ✓✓

Handwritten: $x \oplus y = \overline{xy}$ ✓✓

$x \oplus y = \overline{xy}$

if $xy = 0$ ✓✓

x	y	$x \oplus y$	$x + y$
0	0	0	0
0	1	1	1
1	0	1	1

[MCQ]

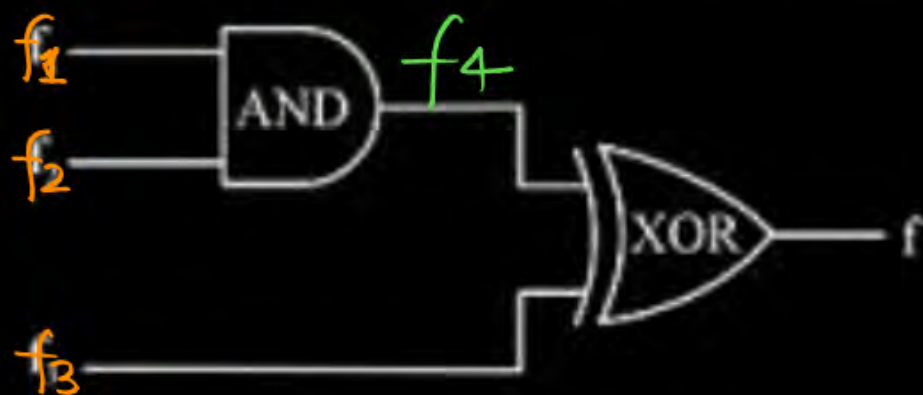


Consider three 4-variable functions f_1 , f_2 and f_3 , which are expressed in sum-of-minterms as

$$f_1 = \Sigma(0, 2, 5, 8, 14), \quad f_2 = \Sigma(2, 3, 6, 8, 14, 15), \quad f_3 = \Sigma(2, 7, 11, 14)$$

For the following circuit with one AND gate and one XOR gate, the output function f can be expressed as:

[GATE-2019-CS: 2M]



- ☐ A $\Sigma(2, 14)$
- ☒ B $\Sigma(7, 8, 11)$
- ☐ C $\Sigma(2, 7, 8, 11, 14)$
- ☐ D $\Sigma(0, 2, 3, 5, 6, 7, 8, 11, 14, 15)$

$$f_4 = \Sigma(2, 8, 14)$$
$$f = \Sigma(7, 8, 11)$$

[MCQ]

Let $\oplus$ and $\odot$ denote the Exclusive OR and Exclusive NOR operations, respectively. Which one of the following is NOT CORRECT? [GATE-2018-CS: 1M]

- A** $P \oplus Q = P \odot Q$
- B** $\bar{P} \oplus Q = P \odot Q$
- C** $P \oplus \bar{Q} = P \oplus Q$
- D** $(P \oplus \bar{P}) \oplus Q = (P \odot \bar{P}) \odot \bar{Q}$

[MCQ]



Let $\oplus$ denote the Exclusive OR (XOR) operation. Let '1' and '0' denote the binary constants. Consider the following Boolean expression for F over two variables P and Q:

$$F(P, Q) = ((1 \oplus P) \oplus (P \oplus Q)) \oplus ((P \oplus Q) \oplus (Q \oplus 0))$$

The equivalent expression for F is

[GATE-2014-CS: 1M] ✓

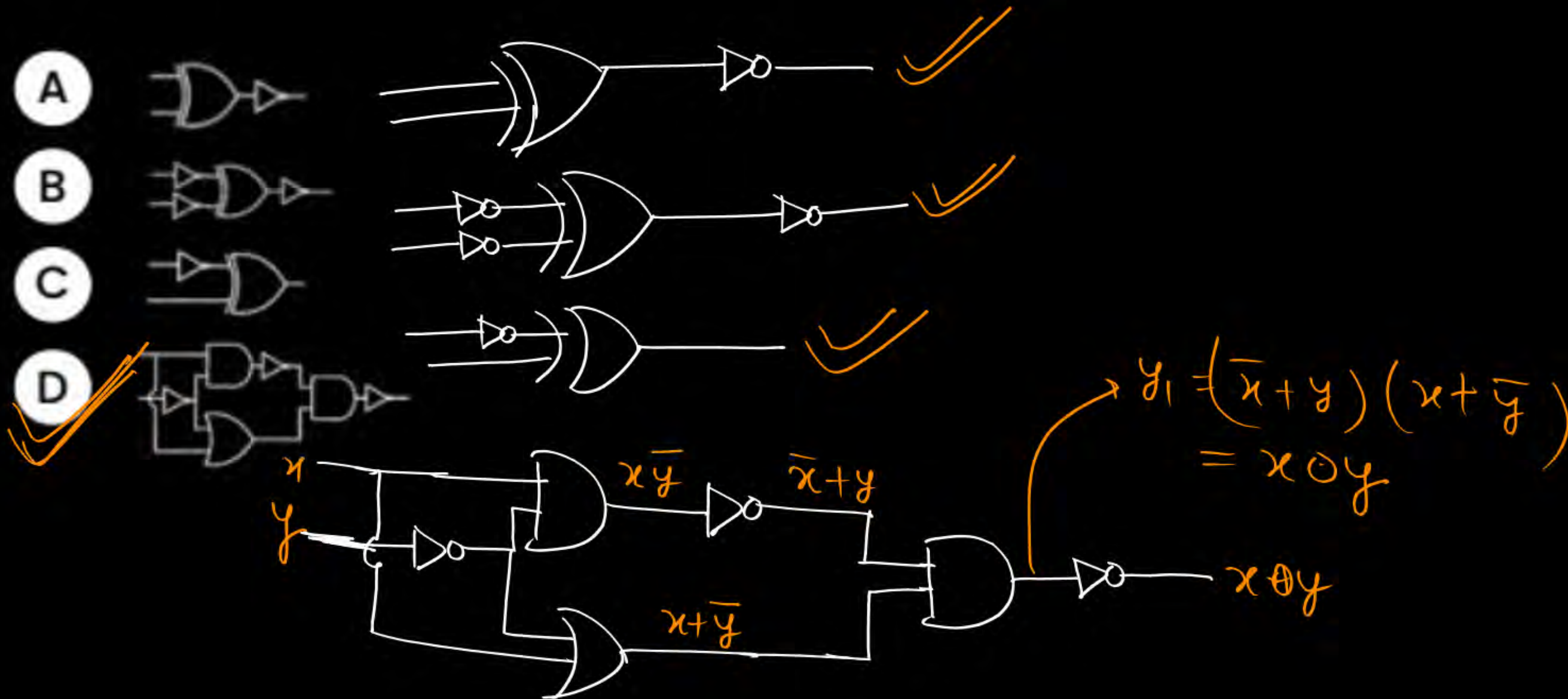
- ☐ A $P + Q$
- ☐ B $(P + Q)'$
- ☐ C $P \oplus Q$
- ☒ D $(P \oplus Q)'$

$$\begin{aligned} P \oplus Q \oplus 1 &= P \oplus \overline{Q} \\ &= P \odot Q \\ &= \overline{P \oplus Q} \end{aligned}$$

[MCQ]

Which one of the following circuits is NOT equivalent to a 2-input X-NOR (exclusive NOR) gate?

[GATE-2011-CS: 1M]



[MCQ]

What is the minimum number of gates required to implement the Boolean function $(AB + C)$ if we have to use only 2-input NOR gates? [GATE-2009-CS: 1M]

$$Y = A \cdot B + C = (A + C) \cdot (B + C)$$

$$\overline{Y} = \overline{A + C} + \overline{B + C}$$

$\downarrow$ ↘
 $\text{1 NOR}(P)$ $\text{1 NOR}(Q)$

$$\overline{Y} = P + Q$$

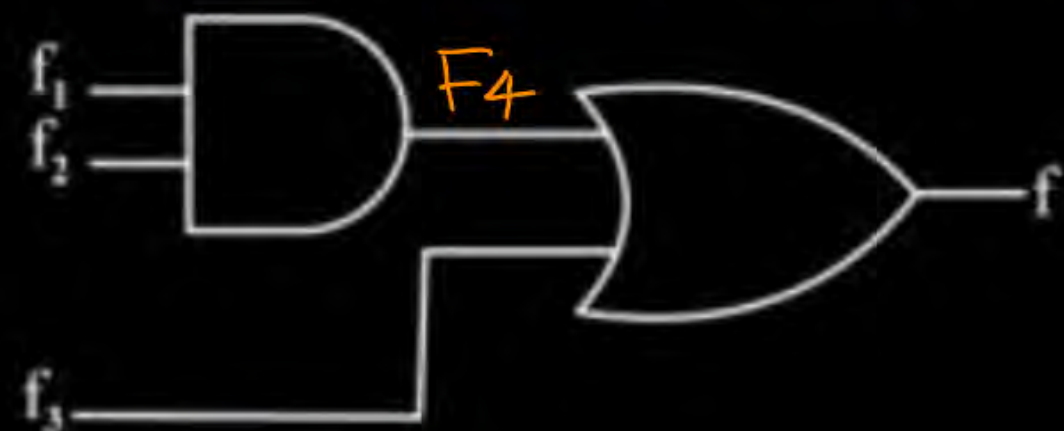
$$Y = \overline{P + Q} \quad (\text{1 NOR})$$

- ☐ A 2
- ☒ B 3
- ☐ C 4
- ☐ D 5

[MCQ]



Given f_1, f_3 and f in canonical sum of products form (in decimal) for the circuit.



[GATE-2008-CS: 1M]

$$F_1 = \sum m(4, 5, 6, 7, 8)$$

$$F_3 = \sum m(1, 6, 15)$$

$$F = \sum m(1, 6, 8, 15)$$

Then f_2 is

☐ A $\sum m(4, 6)$

☐ B $\sum m(4, 8)$

☒ C $\sum m(6, 8)$

☐ D $\sum m(4, 6, 8)$

$$F_1 = \sum (4, 5, 6, 7, 8) \quad F_3 = \sum (1, 6, 15)$$

$$F_2 = ?$$

$$F_4 =$$

$\{ 1, 6, 15 \rightarrow \text{optional}$
 $8 \rightarrow \text{Compulsory}$
other should not be present

$$\sum 6, 8 \quad \sum 1, 6, 9$$

$$\sum 8, 9$$

$f_2 \rightarrow 8 \rightarrow \text{Compulsory}$

$4, 5, 7 \rightarrow \text{should not be there}$

12-terms $\rightarrow$ optional in f_2

$\checkmark \checkmark 12 \checkmark$



[MCQ]

Consider the following Boolean expression for

$$F: F(P, Q, R, S) = PQ + P'QR + P'QR'S$$

The minimal sum-of-products form of F is

[GATE-2014-CS: 1M]

- ☐ A $PQ + QR + QS$
- ☐ B $P + Q + R + S$
- ☐ C $\overline{P} + \overline{Q} + \overline{R} + \overline{S}$
- ☐ D $\overline{P}R + \overline{P}RS + P$

H.W

[Question]



A logical function $f(A, B, C)$ is given as $f(A, B, C) = \overline{A \oplus B \oplus C} + (A \oplus B)\overline{C}$

Then minimum no. of 2-input NAND gate required to implement $f(A, B, C)$ will be _____.

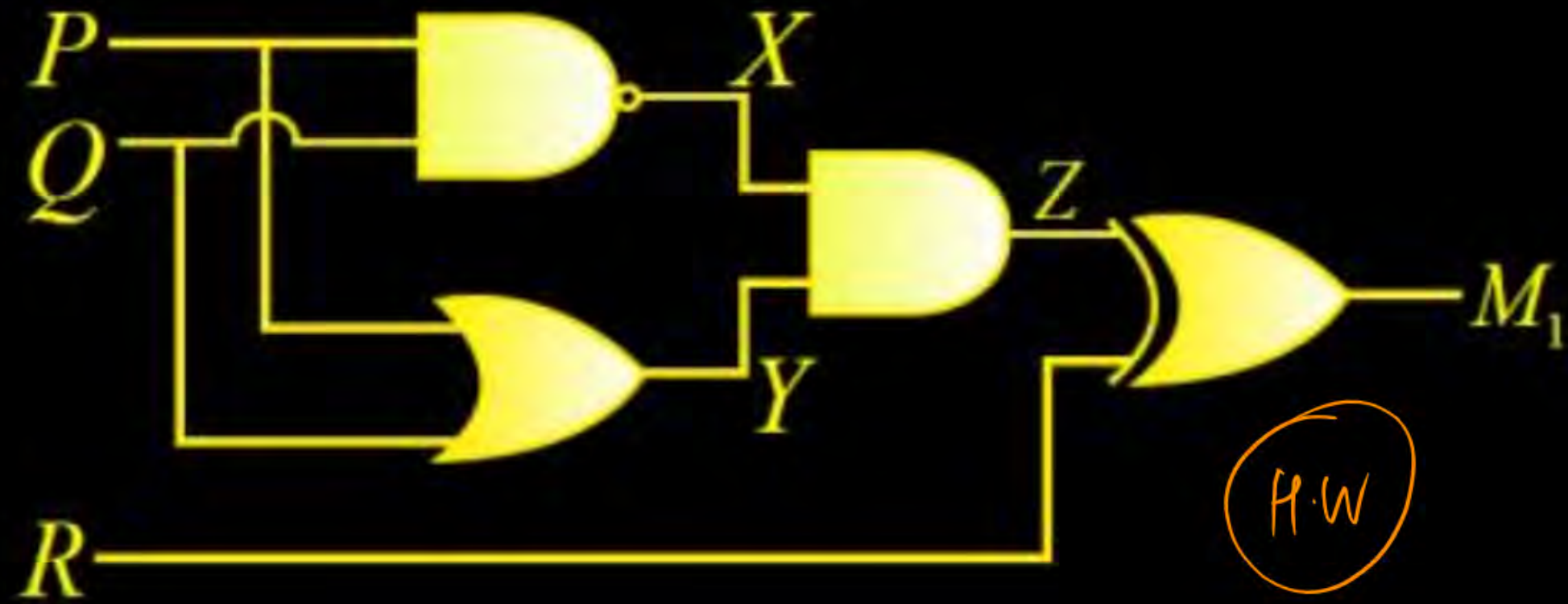
H.W

[Question]



Which of the following Boolean Expression correctly represents the relation between P, Q, R and M_1 ?

- (A) $M_1 = (P \text{ OR } Q) \text{ XOR } R$
- (B) $M_1 = (P \text{ AND } Q) \text{ XOR } R$
- (C) $M_1 = (P \text{ NOR } Q) \text{ XOR } R$
- (D) $M_1 = (P \text{ XOR } Q) \text{ XOR } R$

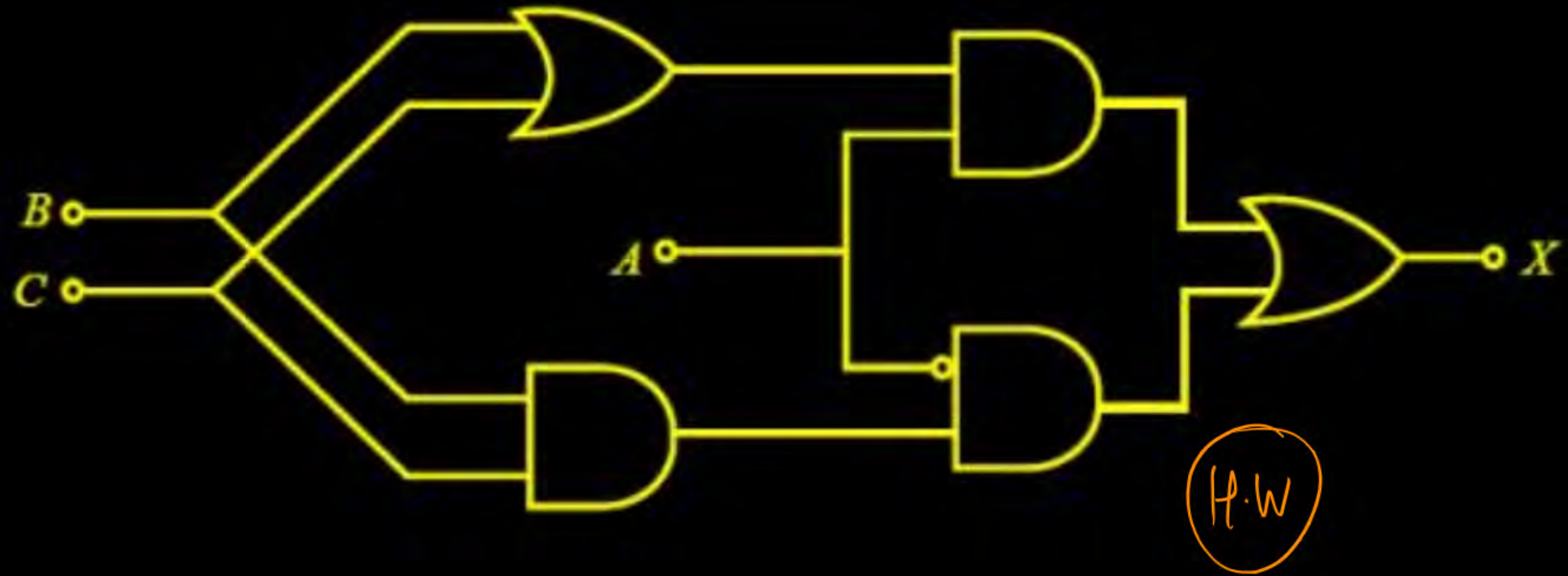


[Question]



In the logic circuit shown in figure, the output X is

- (a) $A\bar{B} + B\bar{C} + C\bar{A}$
- (b) $A + B + C$
- (c) $AB + BC + CA$
- (d) $\overline{AB + BC + CA}$



Thank you

GW
Soldiers !

